

	Answer	Mark
1(a)	The only correct answer is A $\log_{10}$ number of living bacterial cells <i>B is incorrect because the total cell number plotted would not show the death phase</i> <i>C is incorrect because the exponential phase would not be a straight line</i> <i>D is incorrect because the exponential phase would not be a straight line</i>	(1)

Question number	Answer	Additional guidance	Mark
1(b)(i)	An explanation that includes the following points: - because we do not want {other bacteria in the culture / contamination} (1) - as they could {compete for nutrients / produce toxins (to the bacteria being cultured)} OR - because we do not want to transfer the bacteria onto {ourselves / surroundings} - as they may be {pathogenic / harmful} (1)	ACCEPT microorganisms / pathogens ACCEPT microorganisms competition for space ACCEPT prevent infection	(2)

Question number	Answer	Additional guidance	Mark
1(b)(ii)	An explanation that includes the following points: • one named technique explained (1) • a second named technique explained (1)	e.g. working by a Bunsen burner + to create an updraft / to move the microorganisms in the air away / to prevent {bacteria / microorganisms / spores} falling into culture IGNORE kill bacteria in the air autoclaving {agar / culture fluid / equipment} using sterile equipment disinfecting work area washing hands + to kill any {bacteria / microorganisms / spores} present washing hands flaming necks of bottles + to prevent transferring {bacteria / microorganisms / spores} into cultures reducing exposure of cultures to air + to reduce chance of contamination close {doors / windows} + to avoid draughts	(2)